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ELECTROPHYSIOLOGY OF GUINEA-PIG CEREBELLAR NUCLEAR CELLS IN THE *IN VITRO* BRAIN STEM–CEREBELLAR PREPARATION

BY R. LLINÁS AND M. MÜHLETHALER*

From the Department of Physiology and Biophysics, New York University Medical Center, 550 First Avenue, New York, NY 10016, U.S.A.

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SUMMARY

1. Intracellular recordings were obtained from cerebellar nuclear neurones in the isolated brain stem–cerebellar preparation of guinea-pigs *in vitro*. The electrical properties of the cells were quite similar to those reported in *in vitro* slice studies. They had an average resting potential of $-56{\cdot}7 \pm 1{\cdot}8$ mV, an input resistance of $23{\cdot}8 \pm 4{\cdot}9$ MΩ, and a time constant of $12{\cdot}5 \pm 2{\cdot}7$ ms. The action potentials had an average amplitude of $57{\cdot}3 \pm 5{\cdot}28$ mV ($n = 20$).

2. In addition to the ionic mechanisms required for the generation of the fast action potential, cerebellar nuclear neurones displayed a low-threshold Ca^{2+}-dependent spike which produced a powerful rebound excitation following anodal break. This type of electroresponsiveness was absent in the slice preparation.

3. The anodal break response was further enhanced by the presence of a non-inactivating Na^+ conductance similar to that described in Purkinje cells.

4. Following electrical stimulation of the cerebellar cortex or the underlying white matter, excitatory and inhibitory synaptic potentials (EPSP–IPSP sequences) could be recorded in cerebellar nuclear neurones. The EPSPs were elicited by direct activation of collaterals of mossy or climbing fibre afferents. The IPSPs followed direct or orthodromic Purkinje cell activation.

5. The integrity of the olivo-cerebellar system was tested by the administration of harmaline which produced powerful EPSP–IPSP sequences or pure IPSPs in cerebellar nuclear neurones. These IPSPs were often followed by a rebound firing of the cells.

6. These results indicate that the olivo-cerebellar pathway, in addition to its activation of the cerebellar cortex, exerts a powerful and complex set of synaptic events on cerebellar nuclear cells. As such it is a true afferent system, having a distinct role in cerebellar physiology.

INTRODUCTION

A detailed description of the electrophysiological properties of cerebellar nuclear (CN) neurones is one of the fundamental prerequisites to understanding the overall

* Present address: Laboratory of Psychophysiology, CMU, 1 Rue Michel Servet, 1211 Geneva, Switzerland.

function of the cerebellar systems. Indeed, the connectivity of these neurones places them in a truly privileged position. Their axons are the main output of the cerebellum, projecting to the spinal cord, brain stem and forebrain. In turn these neurones are the main target of the Purkinje cell axons, themselves the only output system of the cerebellar cortex. The cerebellar nuclear cells represent, therefore, a hub providing the main functional link between the cerebellar cortex and the rest of the central nervous system.

Previous studies using slices have provided information regarding the electro-responsive properties of the many types of neurones in the cerebellar system (Llinás & Sugimori, 1980*a*, *b*; Gardette, Debono, Dupont & Crepal, 1985*a*, *b*; Gardette & Crepel, 1986; Jahnsen, 1986*a*, *b*) but little insight into the synaptic organization among these cells. By contrast, *in vivo* studies, while elucidating some of the properties of synaptic connectivity (Ito & Yoshida, 1966), have not allowed sufficient recording stability or control of such parameters as ionic environment or drug concentration, to study single-cell electrophysiology in detail.

The present paper characterizes the electrophysiology of CN neurones and reports the finding that the climbing fibre and mossy fibre systems generate, in addition to direct excitatory postsynaptic potentials (EPSPs), distinct patterns of synaptic inhibition in CN neurones via their activation of Purkinje cells. This finding indicates that the two afferents subserve different functions in their own right, as opposed to the view that the olivo-cerebellar system exists mainly as the anatomical basis for cerebellar cortex plasticity (Marr, 1969).

METHODS

The methods used in this study were similar to those described in the previous paper (Llinás & Mühlethaler, 1988). The recordings, to include synaptic potentials and the Ca^{2+}-dependent rebound spike, were made either at 30 or 31 °C or occasionally at 36 °C. Cerebellar nuclear cells were approached from the dorsal surface through the cerebellar cortex (Fig. 1*A*). Using rostral, caudal or parasagittal approaches, recordings were made from cells in all three cerebellar nuclei (i.e. fastigal, interpositus and dentate). The position of the tip of the electrode with respect to the mid-line provided an excellent guide for identifying the particular cerebellar nucleus.

Several criteria were used to identify CN neurones. (a) The profile of spontaneous unitary activity as the electrode was slowly introduced into the cerebellar mass. Thus in a typical microelectrode tract, unit activity was sequentially recorded from Purkinje and granule cell layer neurones. This activity sequence was abruptly interrupted by a zone of electrical silence (generally 1 mm or so thick) as the electrodes passed through the white matter overlying the cerebellar nuclei. Immediately below this region, extracellular single-cell activity was again recorded. These extracellular spikes demonstrated high firing rates and were characterized by a simple spike waveform. (b) The recording depth. The nuclear cell zone of activity was usually found about 3 mm from the cortical surface and was characterized by the homogeneity of the single-cell activity. (c) Intracellular recordings demonstrated antidromic invasion from the cerebellar peduncles, rebound Ca^{2+}-dependent spikes and, following cerebellar white matter stimulation, excitatory postsynaptic potentials (EPSP) followed by large inhibitory synaptic potentials (IPSP); the latter were similar to those reported in the intact cerebellum (Ito & Yoshida, 1964, 1966). (d) Intracellular injection of Lucifer Yellow dye allowed direct visualization of CN neurones.

Intracellular recordings were obtained using either 3 M-potassium acetate- or potassium citrate-filled microelectrodes and standard instrumentation as reported previously (Llinás & Sugimori, 1980*a*; Llinás & Mühlethaler 1988).

Stimulus paradigms. Cerebellar nuclear cells were stimulated electrically through one of an array of five concentric bipolar electrodes (Rhodes Medical Instruments) positioned in the white matter

about 1 mm below the cortical surface. The stimuli were delivered either as single pulses (100–500 μs) or as trains.

Drugs. Pharmacological experiments involved the addition of blocking agents to the intra-arterial perfusate. Tetrodotoxin (TTX, 10^{-6} g/ml) was used to decrease the Na^+ conductance (g_{Na}) and $CaCl_2$ (2·4 mM) was replaced by $CoCl_2$ (2·4 mM) to block the Ca^{2+} conductance (g_{Ca}). Harmaline hydrochloride (Sigma) was used at 5 μg/ml. These substances acted 2–5 min after reaching the arterial system.

TABLE 1. Values for resting potentials, spike height and duration, input resistance and time constants for twenty CN neurones

Cell No.	Resting potential (mV)	Spike height (mV)	Spike duration* (ms)	Input resistance (MΩ)	Time constant (ms)
1	−58	58	0·55	21	13·2
2	−56	57	0·5	23	11·5
3	−58	74	0·55	30	14·3
4	−55	53	0·55	22	16·5
5	−58	54	0·55	26	14·3
6	−60	53	0·55	19	8·8
7	−56	57	0·5	23	11·5
8	−58	58	0·55	21	13·2
9	−55	66	0·44	35	11·0
10	−55	56	0·66	23	8·8
11	−58	55	0·55	30	13·2
12	−55	59	0·66	19	15·4
13	−56	55	0·55	24	12·0
14	−61	64	0·66	30	20·0
15	−58	58	0·55	15	13·2
16	−55	55	0·55	23	12·0
17	−56	53	0·44	25	9·3
18	−55	53	0·44	19	9·9
19	−55	53	0·7	20	11·0
20	−56	55	0·44	29	11·0
Mean ±S.D.	−56·7±1·8	57·3±5·28	0·54±0·077	23·8±4·9	12·5±2·7

*Measured at one-half amplitude.

RESULTS

Intracellular recordings

Activity from individual CN cells was recorded for up to 2 h. The data base in this study comprised 115 intracellularly recorded CN neurones. The electrical properties of twenty cells randomly selected from the above population are summarized in Table 1. The mean and standard deviation for the resting potential of these cells was −56·7±1·8 mV (n = 20). The highest resting potential observed was −61 mV, very much in keeping with values recently reported by Jahnsen (1986*a*) in a study of CN neurones using brain slices. The mean input resistance of the cells included in Table 1 was 23·8±4·9 MΩ (n = 20), while the mean value for the time constant was 12·5±2·7 ms (n = 20). Spontaneous firing was recorded from all impaled CN cells.

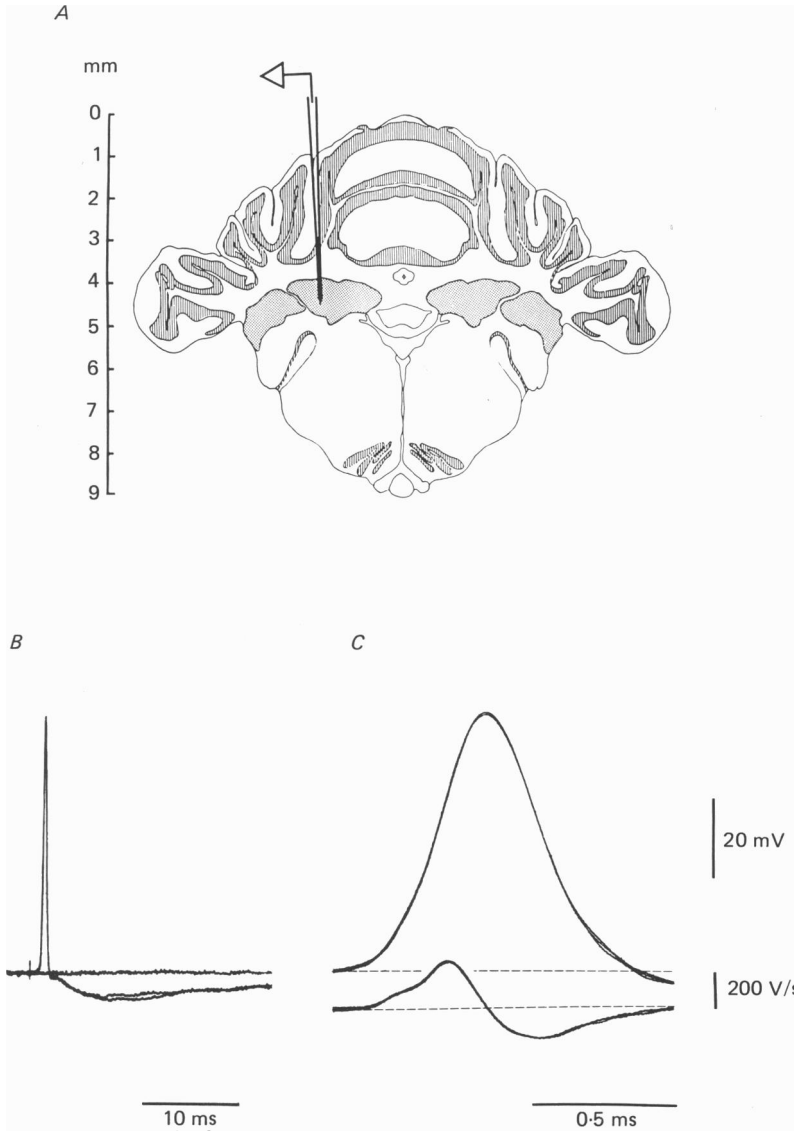

Fig. 1. Electrode path and antidromic action potentials. *A*, drawing of coronal section of the guinea-pig brain stem and cerebellum showing the pathway of the recording microelectrode. Cerebellar nuclei (CN) neurones were usually found 4 mm from the surface immediately beneath the electrically 'silent' white matter. *B* and *C*, action potential elicited by antidromic invasion of a CN neurone shown at slow sweep speed (*B*) and at faster sweep speed (*C*). Lower trace in *C* gives the time derivative of the voltage.

This activity was driven by spontaneous EPSPs and by rebound responses following spontaneously occurring large IPSPs. The level of spontaneous activity varied among cells in a given preparation, the frequency of spontaneous activity fluctuating from several spikes per second to near ten per second.

Direct and antidromic activation

Action potentials elicited by antidromic invasion are illustrated in Fig. 1*B* and *C*. The simple waveform and slight initial segment–somadendritic (IS–SD) break in the rising phase were characteristic of all neurones recorded. The IS–SD break was apparent in the time differential of the potential (lower trace in Fig. 1*C*). The duration, under 0·5 ms at mid-height, was typical, as was their low amplitude (Table 1). In fact, 35% of the spikes overshot the resting potential, 45% had no

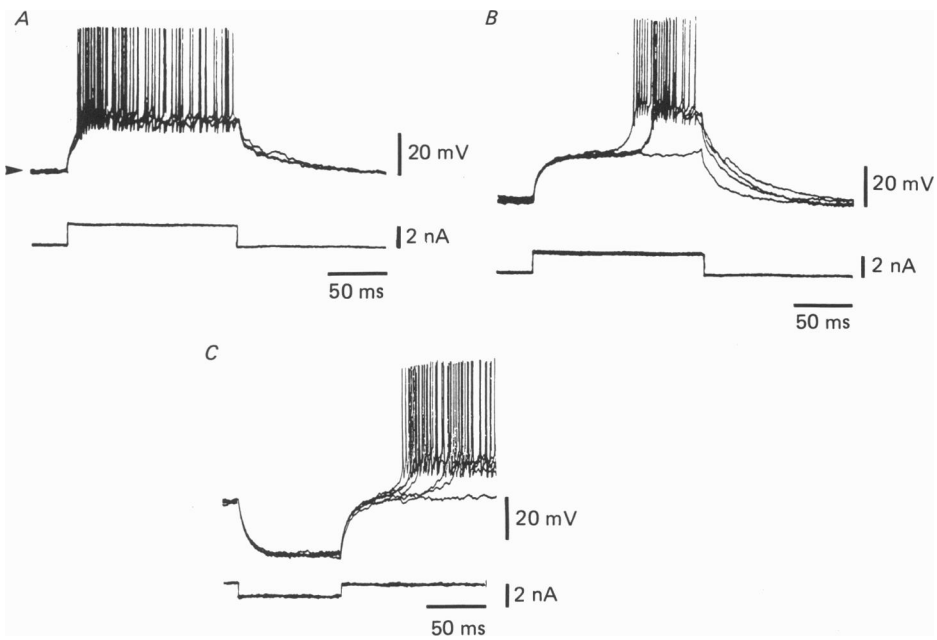

Fig. 2. Firing to current injection. *A*, depolarizing current step delivered at the resting potential (−58 mV) elicited tonic firing. *B*, the same current step as in *A*, delivered from a hyperpolarized membrane level (−72 mV) elicited all-or-none burst responses. *C*, hyperpolarizing current pulses delivered from the resting potential elicited strong rebound responses with variable onsets.

overshoot, and 20% had a spike height equivalent to the resting potential. The low amplitudes of these action potentials were probably related to their rather short duration (Jahnsen, 1986*a*).

Repetitive firing and low-threshold spike burst

Since the repetitive firing properties of cerebellar nuclei neurones in the brain stem–cerebellar preparation were essentially similar to those reported by Jahnsen (1986*a*), they will not be considered here. Rather, we will describe a non-trivial difference between CN cell activity in the slice and the brain stem–cerebellar preparation. In the brain stem–cerebellum the CN neurones demonstrated a clear and rather powerful low-threshold Ca^{2+}-dependent spike which triggered a burst of

fast Na^+-dependent spikes. Indeed, as shown in Fig. 2, the firing properties of CN neurones are markedly modulated by their resting membrane potential. In Fig. 2*A* a nuclear cell was transiently depolarized from its resting potential (−58 mV) by a current pulse which generated repetitive activation of the neurone. In Fig. 2*B* the cell was hyperpolarized with DC injection to −75 mV. Transient current pulses of

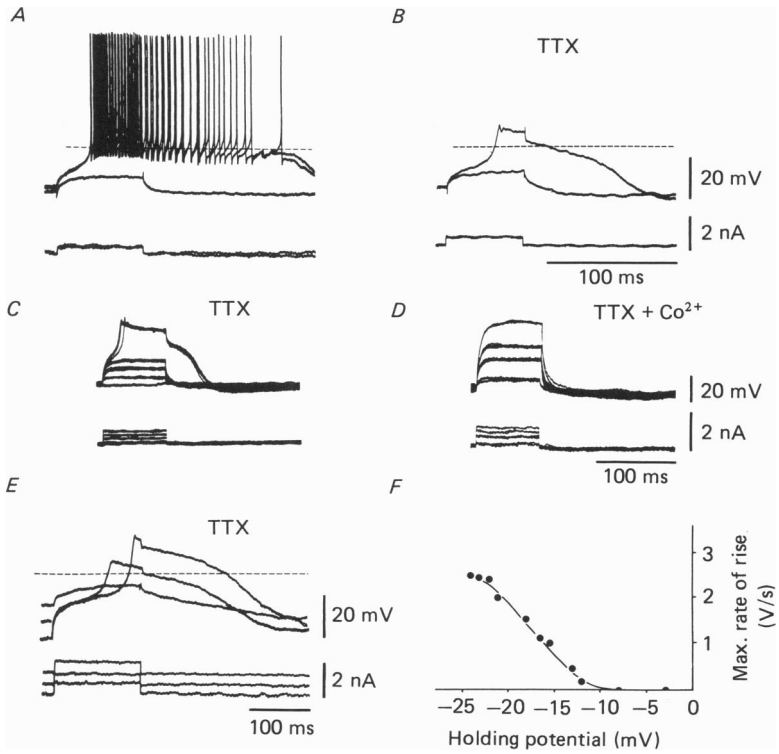

Fig. 3. Ionic mechanisms of burst responses. *A*, prolonged depolarization produced a burst response (two superimposed traces) when the cell was hyperpolarized by 15 mV from rest (dashed line). *B* and *C*, addition of TTX (10^{-6} g/ml) to the perfusate abolished the fast spikes. Note the all-or-none characteristic of the response. *D*, the TTX-insensitive response was abolished after replacing the Ca^{2+} in the perfusate with Co^{2+}. *E*, voltage dependence of de-inactivation of burst response at various holding potentials in the presence of TTX. Note the increased rate of rise of the response at more hyperpolarized levels. *F*, the maximum rate of rise in the presence of TTX is plotted as a function of holding potential for a third cell. Note the characteristic sigmoid shape of this de-inactivation curve.

the same amplitude as those in Fig. 2*A* generated all-or-none, stereotypic spike bursts; this point is particularly clear since the failure of the spike burst in one of the traces serves as a baseline.

Probably more significant to the functioning of the CN neurones is the rebound firing that followed hyperpolarizing current pulses. Indeed, in another cell, when the membrane was hyperpolarized from the resting level (−58 mV) by inward current pulses, rebound bursts were observed at the return of the membrane potential to its

rest value (Fig. 2*C*). This point will be illustrated further in relation to the large CN neurone IPSPs generated by climbing fibre activation of Purkinje cells.

Ionic mechanism for the low-threshold response

The ionic mechanisms underlying the generation of the spike burst were studied pharmacologically utilizing TTX to block voltage-dependent Na^+ channels. In Fig. 3*A* a CN cell is shown to fire a prolonged spike burst which outlasted the stimulus. (The resting potential was maintained at −75 mV with DC injection.) Following addition of TTX (10^{-6} g/ml) to the perfusate (Fig. 3*B*), the fast spikes were blocked, revealing an all-or-none low-amplitude response lasting for nearly 100 ms after the end of the pulse. This TTX-resistant spike showed a clear firing level (Fig. 3*C*) and was eliminated after replacing $CaCl_2$ with $CoCl_2$ in the TTX-containing arterial

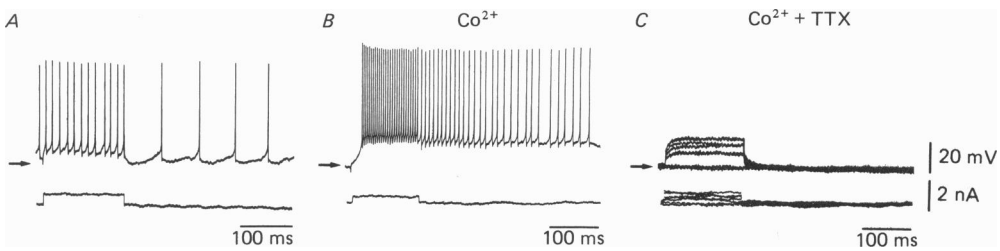

Fig. 4. Non-inactivating Na^+ plateau *A*, at the resting potential depolarizing current pulses generated a train of action potentials. *B*, after replacing the Ca^{2+} in the perfusate with Co^{2+}, a similar current injection elicited a prolonged plateau depolarization. *C*, the response was eliminated by addition of TTX.

perfusion fluid (Fig. 3*D*) indicating that, as in inferior olive neurones (Llinás & Yarom, 1981*a*), the low-threshold spike (LTS) is Ca^{2+} dependent (Llinás & Yarom, 1981*b*). In Fig. 3*E* the voltage dependence of the LTS is illustrated in records obtained from another cell in the presence of TTX (10^{-6} g/ml). A step depolarization was delivered at three membrane potential levels. As the membrane was hyperpolarized from the resting level of −60 mV (dashed line, Fig. 3*E*) the short depolarization elicited different amplitude spikes. The rate of rise of these Ca^{2+}-dependent LTSs is plotted against membrane potential in Fig. 3*F* where 0 mV represents the resting potential (−60 mV). These recordings demonstrate that CN neurones have, in addition to fast, all-or-none Na^+-dependent spikes, a low-threshold Ca^{2+}-dependent conductance. This conductance underlies the generation of LTSs similar to those initially described in inferior olivary neurones (Llinás & Yarom, 1981*a*) and found in other central neurones (Jahnsen & Llinás, 1984*a*; Llinás, Greenfield & Jahnsen, 1984; Carbone & Lux, 1984; Nowycky, Fox & Tsien, 1985).

Non-inactivating Na^+ conductance

Under conditions where the Ca^{2+} conductance was blocked, direct depolarization elicited Na^+-dependent plateau potentials in CN cells. Thus, as shown in Fig. 4*A*, when depolarized from the resting potential (−58 mV, arrow), the cell fired at a high frequency during the intracellular current injection and spontaneously at a lower

frequency after the pulse. When the Ca^{2+}-dependent conductance was blocked by replacing the $CaCl_2$ in the perfusion fluid with $CoCl_2$, a smaller depolarization than in Fig. 4*A* elicited a prolonged plateau potential (Fig. 4*B*). Note also that the after-hyperpolarizations that followed the spikes were significantly decreased after exposure to Co^{2+} (compare Fig. 4*A* and *B*). Addition of TTX to the perfusion fluid blocked both the repetitive firing and the plateau potential (Fig. 4*C*).

Thus, CN neurones have two inward ionic conductances in addition to those underlying the fast action potentials: (i) a non-inactivating Na^+ conductance activated when the membrane is depolarized from the resting level (Fig. 4*B*, and

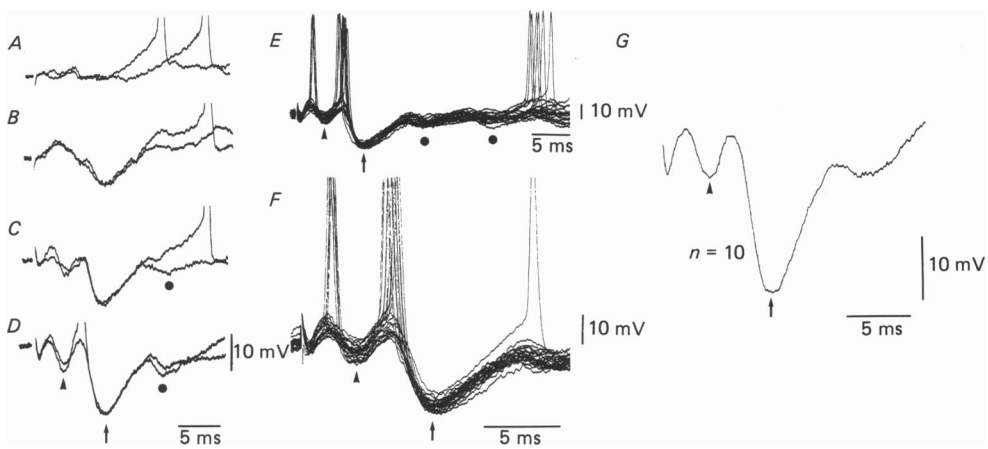

Fig. 5. IPSPs evoked by white matter stimulation. *A–D*, increasing the stimulus strength shows the progressive appearance of two events, an early EPSP–IPSP (arrow-head in *D*), and a late EPSP–IPSP sequence (arrow in *D* and dots in *C* and *D*). *E*, several superimposed responses. *F*, responses in *E* shown at a faster sweep speed. *G*, average of ten responses illustrating the two EPSP–IPSP sequences.

Jahnsen, 1986*b*) and (ii) an inactivating Ca^{2+} conductance which may generate Ca^{2+}-dependent spikes following transient membrane hyperpolarization (Fig. 3*A*). In addition, large depolarizations from the resting potential elicit Ca^{2+}-dependent high-threshold spikes (Jahnsen, 1986*b*).

Synaptic potentials in CN neurones

Electrical stimulation of the cerebellar white matter lying between the cortex and the cerebellar nuclei elicited graded excitatory and inhibitory postsynaptic potentials in CN neurones (Figs 5, 6 and 7). These synaptic potentials can be ascribed to the activation of Purkinje cell axons, mossy and climbing fibre afferents, and the noradrenergic (Hökfelt & Fuxe, 1969) and serotoninergic (Takeuchi, Kimura & Sano, 1982) systems. The conduction velocities of the first three of these are fast and discrete enough to generate well-defined short-latency synaptic potentials. Activation of Purkinje cell axons should generate inhibitory potentials in the CN neurones (Ito *et al.* 1964; Ito, Yoshida, Obata, Kawai & Udo, 1970) and in Deiters' neurones (Ito & Yoshida, 1964, 1966). On the other hand, since mossy and climbing

fibres send collaterals to the cerebellar nuclei on their way to the cortex (Bloedel & Courville, 1981), excitatory postsynaptic potentials should be observed in CN neurones following white matter stimulation.

Indeed, as illustrated in Fig. 5, the response of CN cells to white matter stimulation was often characterized by an early, graded EPSP which was invariably followed by

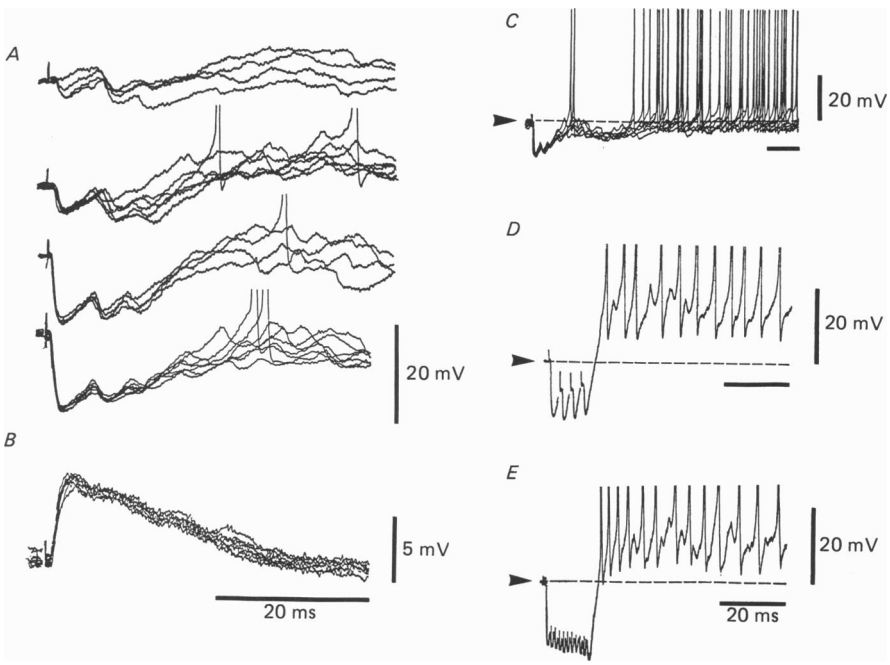

Fig. 6. Summing and reversal properties of the cerebellar nucleus IPSPs. *A*, four to six responses are superimposed for each of four stimulus strengths (1·2, 1·4, 1·6 and 2 times threshold) to show the reliability of the multiple IPSPs. *B*, a 2 times threshold stimulus delivered from a hyperpolarized membrane potential reversed the sign of the IPSPs and did not change their repetition rate. *C*, a single stimulus (2 times threshold) generated a triple IPSP and a rebound spike response followed by a second inhibition and rebound. *D* and *E*, repeated stimuli at 200/s (*D*) or 500/s (*E*) delivered from rest show summation of the IPSPs and a rebound burst of spikes.

a short-lasting IPSP (arrow-head in Fig. 5*D–G*). An example of the graded nature of these responses is shown in Fig. 5*A–D*. The initial EPSP, because of its short latency (0·3–0·5 ms), was ascribed to direct activation of mossy fibre collaterals. The IPSP latency of 1·5–2·5 ms is in agreement with the activation time for Purkinje cell axons, as initially suggested by Ito & Yoshida (1966). A second EPSP–IPSP sequence was generally recorded with a latency of 3·0–3·5 ms, well within the conduction velocity of the climbing fibre afferent system (Eccles, Llinás & Sasaki, 1966). The second graded EPSP was followed by an IPSP (arrow in Fig. 5*D–G*) with a latency of 4–6·5 ms. This large inhibitory potential was followed by two or three smaller IPSPs (Fig. 5*C–E*, dots), and by action potentials indicating that the hyperpolarization

produced by the IPSP was sufficient to activate the rebound response, although the IPSPs were evoked from the resting potential. An average of ten responses is shown at a higher gain and sweep speed in Fig. 5*G*. This illustrates the timing of the double EPSP–IPSP sequence. The latencies of the different synaptic events and their time course correspond quite closely to *in vivo* results obtained under similar conditions (Eccles, Ito & Szentagothai, 1967).

An example in which IPSPs were produced by a single white matter stimulus in the absence of the early EPSP is shown in Fig. 6. A set of superimposed traces illustrate responses elicited by white matter stimulation at four stimulus strengths (1·2, 1·4, 1·6 and 2 times threshold). Note the regularity in the onset and amplitude of the initial IPSP and of the IPSPs that follow. That these potentials are IPSPs was demonstrated by hyperpolarizing the cell by 25 mV (to −84 mV) (Fig. 6*B*), at which level a stimulus two times threshold elicited a clearly reversed initial IPSP. The subsequent synaptic potentials were also reversed (Fig. 6*B*). The response of this cell to white matter stimulation from resting potential is illustrated at a slow sweep speed and low gain in Fig. 6*C*. Note that the single stimulus generated a rebound response which generated action potentials. This was followed by a secondary period of inhibition which is probably a continuation of the early hyperpolarization which was separated from the initial response by the rebound spike activity. After these transient responses the cell continued to fire at its normal spontaneous rate.

The issue of the rate at which the Purkinje cell IPSPs can follow electrical stimuli without failing was addressed by delivering trains of stimuli through the white matter electrode. Four stimuli, delivered at 5 ms intervals, each elicited an IPSP, and a strong rebound potential followed the stimulus train (Fig. 6*D*). The maximum membrane hyperpolarization attained by the IPSPs was 18 mV, bringing the membrane potential to −76 mV. A similar set of stimuli delivered at 2 ms intervals produced IPSP summation to a hyperpolarizing amplitude of 20 mV (Fig. 6*E*). The question of IPSP summation is significant because climbing fibre activation of Purkinje cells is known to generate spike trains having interspike intervals as short as 2 ms (Eccles, Llinás & Sasaki, 1968). The present records indicate that quite effective inhibitory synaptic summation can occur in CN neurones at that frequency.

The issue of on-going repeated activation of CN neurones was also addressed in the brain stem–cerebellar preparation. Repeated activation of IPSPs following a single stimuli could be quite dramatic, as shown in Fig. 7. Here a single white matter stimulus was followed by a set of three well-defined IPSPs (dots), and then by a rhythmic set of IPSPs lasting for hundreds of milliseconds (Fig. 7*A* and *B*). Two points seem clear from these data. First, white matter stimulation seems to reset the rhythmic inhibitory input to the CN cells. For example, in Fig. 7*B* the IPSPs in seven superimposed traces have an almost identical frequency and timing. Note also that the IPSPs occurred with an interval of 12 ms and were followed by clear rebound responses, often reaching the firing level. The uniformity of these rhythmic IPSPs was sufficient to allow the averaging of ten successive responses without degradation (Fig. 7*C*). An autocorrelogram function of this average (Fig. 7*D*) shows a primary interval between IPSPs of 12 ms (83·3 Hz). Furthermore, analysis of IPSPs occurring several seconds after the stimulus demonstrated a similar

rhythmicity indicating that such repeated membrane potentials are independent of the autorhythmic properties of CN cells (Fig. 8*A* and *B*). That such events are IPSPs was demonstrated by their reversal during injection of hyperpolarizing current as shown in Fig. 7*E* and *F* where a reversal of the initial IPSP as well as those potentials that follow was obtained at a membrane potential 25 mV negative with respect to

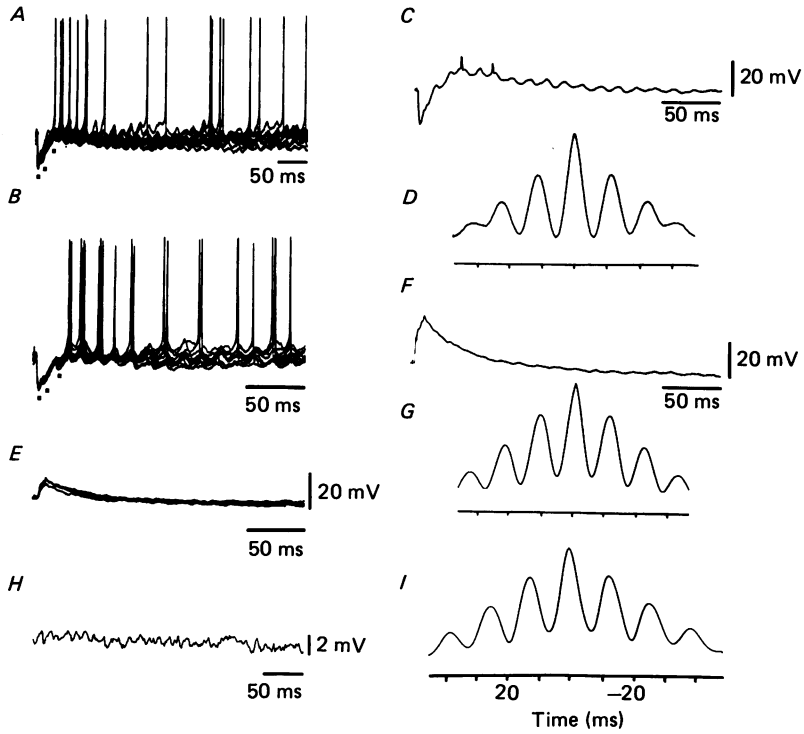

Fig. 7. Oscillatory IPSPs following white matter stimulation. *A*, single stimulus to the cerebellar cortex elicited a large IPSP followed by rebound firing of the cell and a series of smaller, rhythmic IPSPs. *B*, a similar response to that in *A* is shown at a faster sweep speed. *C* and *D*, the regularity of the small IPSPs is illustrated by the averaged responses in *C* ($n = 10$) and the autocorrelation function has a dominant frequency near 83 Hz (*D*). *E–G*, the IPSPs were reversed in sign by hyperpolarizing DC injection (*E*), but the rhythmicity of the synaptic potentials persisted (*F*) as indicated by the autocorrelation function (*G*). *H* and *I*, spontaneous oscillations (*H*) in the same neurone show a similar regularity (*I*) having the same dominant frequency as the evoked IPSP; compare autocorrelograms in *D*, *G* and *I*.

the resting potential. This did not affect the rhythmicity of the IPSPs, as indicated by the autocorrelogram in Fig. 7*G*.

The rhythmicity following the evoked IPSP was present spontaneously in the same CN cell (Fig. 7*H* and *I*). The autocorrelogram of the data in Fig. 7*H* shown in *I* illustrated once again the clear 12 ms rhythm.

Effects of harmaline

A quite different tool for the study of the olivo-cerebellar system is afforded by the utilization of the alkaloid harmaline. Harmaline was used in these studies because it is a well-known tremulant acting quite specifically on the neurones in the inferior olivary nucleus (Llinás & Volkind, 1973; de Montigny & Lamarre, 1974; Llinás & Yarom, 1986). This was confirmed in the brain stem–cerebellum preparation, as

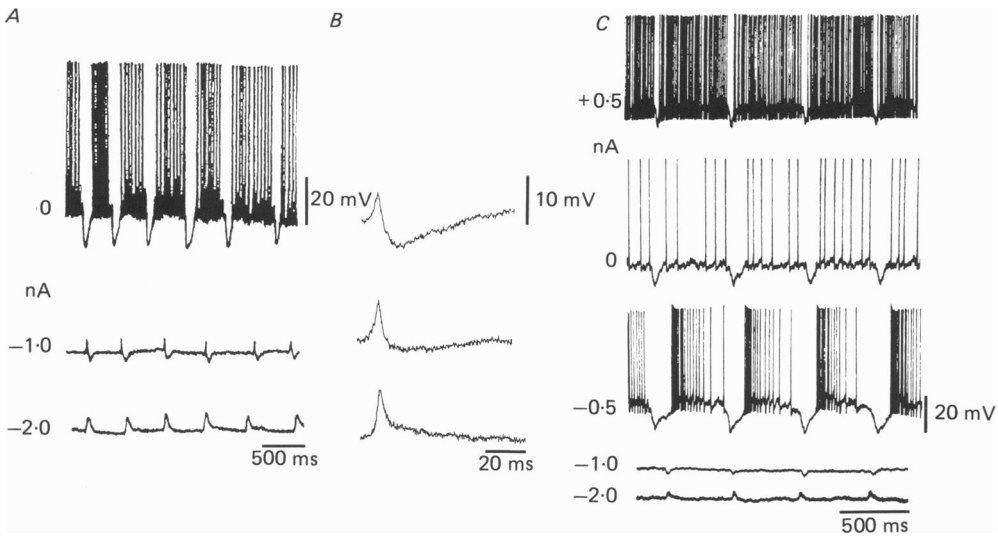

Fig. 8. The effect of harmaline. *A*, addition of 5 μg/ml of harmaline to the perfusate generated large-amplitude EPSP–IPSP sequences (0 nA). With DC hyperpolarization of −1·0 nA, the EPSP–IPSP sequence can be observed in the absence of action potentials; with −2·0 nA hyperpolarization the IPSPs reversed. *B*, single EPSP–IPSP sequences are shown at rest and with DC hyperpolarizations of −1·0 and −2·0 nA. *C*, rhythmic IPSPs at resting potential (0 nA). With a 0·5 nA depolarization the cell fired repetitively and was inhibited by the rhythmic IPSP. With a 0·5 nA hyperpolarization, the IPSPs were followed by a rebound spike burst. At 1 nA hyperpolarization, the rebound potential disappeared. Further hyperpolarization (−2 nA) produced a reversal of the IPSPs.

transections of the inferior cerebellar peduncle (which interrupts the olivo-cerebellar input to the ipsilateral cerebellar nuclei) abolished the climbing fibre responses.

The effect of harmaline on the brain stem–cerebellar preparation as recorded from the cerebellar nucleus is illustrated in Fig. 8. Basically, two kinds of electrophysiological behaviour were recorded following addition of harmaline to the perfusate (5 mg/ml). The first consisted of an early synaptic excitation followed by a powerful synaptic inhibition. An example of this type of response is illustrated in Fig. 8*A* in a CN neurone that was spontaneously active prior to harmaline administration. Five minutes after perfusion with this alkaloid, activation of the EPSP–IPSP sequence was clearly observable, having a frequency of approximately 2/s. In the first trace in Fig. 8*A* (0 nA) the EPSP–IPSP sequences were strong enough to activate the neurone to the firing level during the excitatory phase and to

totally inhibit firing during the IPSPs. When the membrane potential was hyperpolarized by a 1 nA DC injection, the EPSP–IPSP sequence was clearly observed in the absence of spike activation (Fig. 8*A*, −1·0 nA). Further hyperpolarization produced a reversal of the IPSP with little change in the time course of the EPSP (Fig. 8*A*, −2 nA). Single recordings of synaptic sequences are illustrated in Fig. 8*B* for 0, −1·0 and −2·0 nA DC injection. These records indicate that harmaline may produce the activation of CN cells through the activation of the olivo-cerebellar collaterals, which elicit direct EPSPs, or through activation of a rebound response which follows Purkinje cell-mediated IPSPs (Fig. 8*A*, 0 nA).

The second type of synaptic pattern generated by harmaline is characterized by an IPSP in the absence of the early EPSP. Such a pattern is shown in Fig. 8*C*, where the rhythm of the IPSPs is clearly defined and is shown to be independent of membrane potential. Indeed, when the cell was depolarized such that continuous firing was generated (Fig. 8*C*, +0·5 nA), a clear rhythmic inhibition could be observed at the same frequency as that seen at rest (Fig. 8*C*, 0 nA). As the cell was hyperpolarized from rest, it first demonstrated the expected increase in the low-threshold rebound response (Fig. 8*C*, −0·5 nA). With further hyperpolarization, a reduction of the IPSP amplitude (Fig. 8*A*, −1 nA), and finally a reversal (Fig. 8*C*, −2·0 nA), were obtained.

DISCUSSION

The electrical properties of CN neurones in the brain stem–cerebellar preparation resemble those of thalamic cells studied in brain slices. Indeed, as in thalamic cells (Llinás & Jahnsen, 1982; Jahnsen & Llinás, 1984*a*, *b*), and in agreement with recent studies by Jahnsen (1986*a*, *b*), CN cells were found to have a non-inactivating Na^+ conductance similar to that described in Purkinje cells (Llinás & Sugimori, 1980*b*), in addition to the usual Na^+- and K^+-dependent conductances which generate fast action potentials (Hodgkin & Huxley, 1952). In contrast to the results of Jahnsen (1986*a*), however, a powerful low-threshold spike (LTS) was also identified which makes the similarity between CN and thalamic cells more striking. The presence of a LTS is probably of major functional significance in these neurones as such bursts can easily be activated following the powerful inhibitory postsynaptic potentials produced by the climbing fibre activation of Purkinje cells.

The low-threshold calcium spike and its rebound property

The experimental results reported here demonstrate that, as in neurones of the inferior olive (Llinás & Yarom, 1981*b*), thalamus (Jahnsen & Llinás, 1984*b*), substantia nigra (Llinás *et al.* 1984), dorsal root ganglion (Carbone & Lux, 1984; Nowycky *et al.* 1985) and spinal cord (Murase & Randic, 1983), activation of LTSs is Ca^{2+} dependent. Indeed, the voltage dependence of this response, its insensitivity to TTX and its sensitivity to Co^{2+} are in agreement with the presence of a conductance similar to that identified in the cells enumerated above. By analogy with those cells then we may suggest that this conductance is located in the somatic membrane.

The presence of a LTS is, on the other hand, surprising as it is absent in the cerebellar slice (Jahnsen, 1986*a*). This indicates that the electrical properties of CN

neurones connected to their afferents are not the same as those in the 'deafferented' slice preparation. The mechanism by which the presence of a low-threshold Ca^{2+} conductance could be modulated, however, is a matter of conjecture. It must be considered that in the slice some of the catecholaminergic (Hökfelt & Fuxe, 1969) and serotoninergic (Takeuchi *et al.* 1982) pathways may be damaged, and thus that the difference between our finding and those of Jahnsen may be related to the lack of such modulatory properties in the slices. This argument is also relevant when considering under what conditions the LTS is likely to be generated in cerebellar nuclear neurones.

Synaptic organization of cerebellar nuclear neurones

In agreement with, and following a paradigm similar to, that used *in vivo* by Ito & Yoshida (1966), electrical stimulation of the white matter in different folia elicited an early excitatory synaptic potential followed by a short inhibition (Fig. 5). Also, in agreement with the *in vivo* studies, activation of the early excitatory synaptic potentials was achieved by stimulation of more restricted regions of white matter than were the inhibitory potentials. The early IPSP appears to be due to direct activation of the Purkinje cell axons, while the early excitation probably represents antidromic activation of cortical afferent collaterals. The second excitatory potential observed under these stimulation conditions (Fig. 5), and the powerful inhibition that followed, suggests that the second wave of activity is generated via the climbing fibre system. This is supported by the observations that: (a) activation of the inferior olive by harmaline can generate EPSP–IPSP sequences and IPSPs without the initial EPSP and (b) these harmaline-mediated climbing fibre responses were abolished by section of the inferior peduncle. Regarding the spontaneous activity, at the CN level, background inhibitory potentials indicate that Purkinje cells fire with some degree of synchronicity.

The rhythmic inhibitory potentials observed after harmaline administration are produced by the synchronous activation of inferior olivary neurones in the brain stem–cerebellar preparation. However, harmaline generates a rhythmic firing at a frequency of 2/s as opposed to 10/s which is seen *in vivo* (Llinás & Volkind, 1973; de Montigny & Lamarre, 1974). The reason for the low firing frequency is not clear, but does resemble the frequency of oscillations recorded at 36 °C in inferior olivary slices (Llinás & Yarom, 1986).

Time course of the harmaline EPSP–IPSP sequence and of the IPSPs

The results illustrated in Fig. 8 indicate that harmaline can produce two types of responses in CN neurones in this preparation: either an EPSP–IPSP sequence or a simple IPSP. In the first case the time course for the sequence indicates that the early synaptic potentials are most probably due to the close-to-synchronous activation of the olivo-cerebellar system which sends collateral synaptic input to the CN on its way to the cerebellar cortex. The EPSP had a time to peak of 3·5–5 ms and an overall duration of 10–12 ms. The IPSPs had a time to peak of 20–25 ms, and lasted 80–100 ms. When the initial EPSP was absent, the duration of the IPSP was quite similar to that when the early excitatory synaptic potentials were present. Thus, as

shown in Fig. 8*C*, the IPSPs recorded at the resting membrane potential had an overall duration of close to 100 ms reaching a peak at approximately 20 ms.

These results suggest that the synaptic potentials are generated by a close-to-synchronous activation of the olivo-cerebellar afferent system where the time dispersion of the afferent volley may be in the order of 3–5 ms, judging from the time course of the early EPSP. In addition, the distribution of inhibitory potentials over the cerebellar nucleus seems to be wider than the area of EPSP–IPSP innervation. This in turn suggests that the distribution of the climbing fibre collateral system over the cerebellar nuclei is probably more restricted than that of the climbing fibre system over the cerebellar cortex.

The functional significance of the electrical properties of cerebellar nuclear cells in the context of the cortico-nuclear projection

The results described here suggest that CN neurones are regulated in three ways: (1) By excitatory input from collaterals of the cerebellar afferents. (2) By inhibitory inputs from Purkinje cells generated by mossy and climbing fibre activation. Mossy fibre activation should generate a tonic inhibition due to simple spikes in Purkinje cells firing disynchronously at 30–40 Hz (Bell & Grimm, 1969), as well as transient inhibition following a specific mossy fibre volley. (3) By the climbing fibre system which originates in the inferior olivary nucleus where the neurones are known to have powerful pacemaker properties and to be electrotonically coupled (Llinás & Yarom, 1986). This system generates rhythmic and synchronous activation of Purkinje cells, and thus not surprisingly the inhibition generated via these afferents on CN neurones has similar rhythmic properties.

In this third type of CN regulation the membrane properties of CN neurones become particularly distinct since the punctate and rather powerful synaptic EPSP–IPSP sequences are often followed by a rebound spike burst. This means then, that if a sufficient number of inferior olivary neurones, having a common rhythmicity, are activated synchronously at any particular time, a large and equally synchronous activation of Purkinje cells will occur. Such activation would be reflected in the CN as inhibition followed by a rebound activation which would be transmitted to the rest of the nervous system by the axons of the CN cells. That this is probably also the case *in vivo* is indicated by multiple-electrode recording studies of Purkinje cell activity in the rat cortex *in vivo* (Bower & Llinás, 1983). The climbing fibre system generates rostro-caudal bands of synchronous Purkinje cell activation and Purkinje cells within such bands (which are about 200 μm wide) fire with the same frequency, and at intervals as close as 0·5–1 ms, indicating that the olivo-cerebellar system is quite uniquely organized (Sasaki & Llinás, 1985).

The question as to the level of membrane potential that must be attained by the cerebellar nuclear neurone to generate rebound spike bursts must remain moot at this point. Such rebounds may be observed from resting membrane potential (Figs 5–8) and become quite powerful as the cells are hyperpolarized. However, the finding that such rebound spikes are absent in the cerebellar slice does indicate that the low-threshold conductance in CN neurones may be modulated by parameters other than membrane potential.

This sequence of events after harmaline administration is illustrated by the diagram in Fig. 9. If the activity of the circuit is taken to start at the inferior olive, the axons of these cells may be followed to the cortex as climbing fibres or to the CN as collaterals. The activation of this pathway generates activation of Purkinje cells and of CN neurones. The system generates the second EPSP–IPSP sequence also observed in this preparation with a latency of 3–4·5 ms and the large IPSP observed

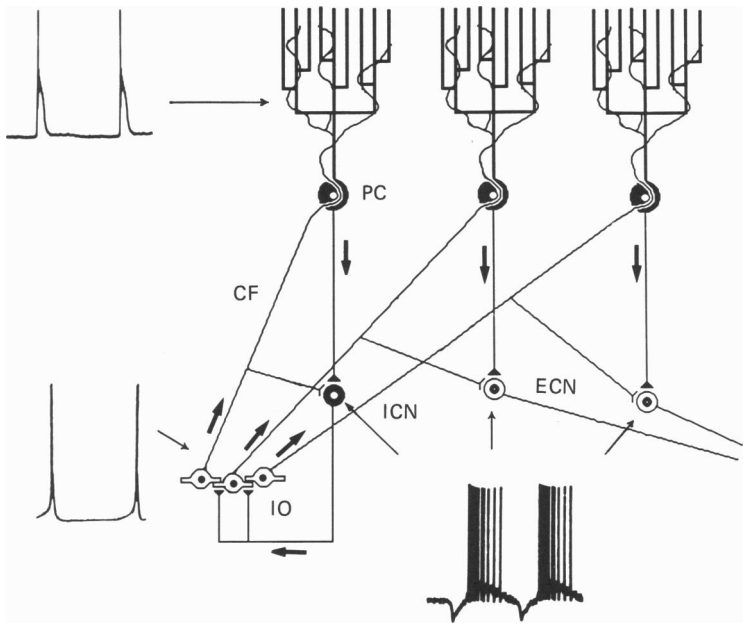

Fig. 9. Diagram of the circuit involved in the production of rhythmic activity in the olivo-cerebellar system in the presence of harmaline. The rhythmic activity of the inferior olivary neurones (IO, lower left) is synchronized by the presence of gap junctions (illustrated by the overlapping dendrites). This activity elicits rhythmic and synchronous climbing fibre (CF) activation of Purkinje cells (PC, upper left) and collateral activation of inhibitory cerebellar neurones (ICN) and excitatory cerebellar neurones (ECN). The activation of Purkinje cells generates powerful inhibitory potentials in the CN neurones which are followed by a rebound activation (bottom right). The activation of excitatory nuclear neurones is transmitted to the rest of the CNs via the cerebellar peduncles. The activation of the inhibitory neurones modulates the excitability of the inferior olivary cells via the GABAergic nucleo-olivary pathway that terminates in the inferior olive glomerulus.

after harmaline administration. These IPSPs are then followed by a powerful rebound excitation which is in turn transmitted to the rest of the CNS as an excitatory input and to the inferior olive as an inhibitory input (via the GABAergic nucleo-olivary pathway) (Mugnaini & Oertel, 1981; Sotelo, Gotow & Wassef, 1986). This pathway has been suggested to have a decoupling effect on the electrotonic junctions in the inferior olive (Sasaki & Llinás, 1985; Sotelo *et al.* 1986). In short, the olivo-cerebellar system would serve as an oscillatory circuit capable of generating timing sequences such as observed in tremor and in the organization of co-ordinated movements (cf. Llinás, 1985).

From this point of view the electroresponsive properties of CN cells seem particularly well-suited to serve in the double role of (1) relaying nuclei for the mossy fibre–nuclear and the mossy fibre–parallel fibre–Purkinje cell–nuclear inhibitory pathway and (2) as a timing device subserved mainly by the olivo-nuclear and olivo-Purkinje cell–nuclear pathways. Further, such tremor is produced by a synchronous and well-defined activity which may be recorded in the cerebellar nuclei, in reticulo-spinal neurones, and in motoneurones (Llinás & Volkind, 1973). The present results are in contrast with the view that the sole role of the climbing fibre afferent is that of modulating the alleged long-term plasticity of the parallel fibre to Purkinje cell synapse. This has been theoretically assumed by some to be at the basis of a possible learning property by the cerebellar cortex (Marr, 1969). The results here indicate that the climbing fibre system has a powerful effect on the CN cells via the Purkinje cells. As such it probably has a basic function in its own right and thus a more fundamental role than being a modulator of the parallel fibre–Purkinje cell synapse.

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REFERENCES

Bell, C. C. & Grimm, R. J. (1969). Discharge properties of Purkinje cells recorded on single and double microelectrodes. *Journal of Neurophysiology* **32**, 1044–1055.

Bloedel, J. R. & Courville, J. (1981). *Cerebellar Afferent Systems*, chap. 16, *Handbook of Physiology*, vol. II, *The Nervous System*, part II, ed. Brooks, V. B., pp. 735–829. Bethesda: American Physiological Society.

Bower, J. & Llinás, R. (1983). Simultaneous sampling of the responses of multiple, closely adjacent, Purkinje cells responding to climbing fibre activation. *Society of Neuroscience Abstracts* **9**, 607.

Carbone, E. & Lux, H. D. (1984). A low voltage-activated calcium conductance in embryonic chick sensory neurons. *Biophysical Journal* **46**, 413–418.

de Montigny, C. & Lamarre, Y. (1974). Rhythmic activity induced by harmaline in the olivo-cerebellar–bulbar systems of the cat. *Brain Research* **53**, 81–95.

Eccles, J. C., Ito, M. & Szentagothai, J. (1967). *The Cerebellum as a Neuronal Machine*. Berlin: Springer.

Eccles, J. C., Llinás, R. & Sasaki, K. (1966). The excitatory synaptic actions of climbing fibres on the Purkinje cells of the cerebellum. *Journal of Physiology* **182**, 268–296.

Gardette, R. & Crepel, F. (1986). Chemoresponsiveness of intracerebellar nuclei neurones to L-aspartate, L-glutamate and related derivatives in rat cerebellar slices maintained *in vitro*. *Neuroscience* **18**, 93–103.

Gardette, R., Debono, M., Dupont, J. L. & Crepel, F. (1985*a*). Electrophysiological studies on the postnatal development of intracerebellar nuclei neurones in rat cerebellar slices maintained *in vitro*. I. Postsynaptic potentials. *Developmental Brain Research* **19**, 47–55.

Gardette, R., Debono, M., Dupont, J. L. & Crepel, F. (1985*b*). Electrophysiological studies of the postnatal development of intracerebellar nuclei neurones in rat cerebellar slices maintained *in vitro*. II. Membrane conductances. *Developmental Brain Research* **20**, 97–106.

Hodgkin, A. L. & Huxley, A. F. (1952). The components of membrane conductance in the giant axon of *Loligo*. *Journal of Physiology* **116**, 473–496.

Hökfelt, T. & Fuxe, K. (1969). Cerebellar monoamine nerve terminals: a new type of afferent fiber to the cortex cerebelli. *Experimental Brain Research* **9**, 63–72.

Ito, M. & Yoshida, M. (1964). The cerebellar evoked monosynaptic inhibition of Deiters' neurones. *Experientia* **20**, 515–516.

Ito, M. & Yoshida, M. (1966). The origin of cerebellar-induced inhibition of Deiters' neurones. I. Monosynaptic initiation of the inhibitory postsynaptic potentials. *Experimental Brain Research* **2**, 330–349.

ITO, M., YOSHIDA, M. & OBATA, K. (1964). Monosynaptic inhibition of the intracerebellar nuclei induced from the cerebellar cortex. *Experientia* **20**, 575–576.

ITO, M., YOSHIDA, M., OBATA, K., KAWAI, N. & UDO, M. (1970). Inhibitory control of intracerebellar nuclei by the Purkinje cell axons. *Experimental Brain Research* **10**, 64–80.

JAHNSEN, H. (1986*a*). Electrophysiological characteristics of neurones in the guinea-pig deep cerebellar nuclei *in vitro*. *Journal of Physiology* **372**, 129–147.

JAHNSEN, H. (1986*b*). Extracellular activation and membrane conductances of neurones in the guinea-pig deep cerebellar nuclei *in vitro*. *Journal of Physiology* **372**, 149–168.

JAHNSEN, H. & LLINÁS, R. (1984*a*). Electrophysiological properties of guinea-pig thalamic neurones: an *in vitro* study. *Journal of Physiology* **349**, 205–226.

JAHNSEN, H. & LLINÁS, R. (1984*b*). Ionic basis for the electroresponsiveness and oscillatory properties of guinea-pig thalamic neurones *in vitro*. *Journal of Physiology* **349**, 227–247.

LLINÁS, R. (1985). Rebound excitation as the physiological basis for tremor: a biophysical study of the oscillatory properties of mammalian central neurones *in vitro*. In *Movement Disorders: Tremor*, ed. FINDLEY, L. J. & CAPILDEO, R., pp. 165–182. London: Macmillan.

LLINÁS, R., GREENFIELD, S. A. & JAHNSEN, H. (1984). Electrophysiology of pars compacta cells in the *in vitro* substantia nigra. A possible mechanism for dendritic release. *Brain Research* **294**, 127–132.

LLINÁS, R. & JAHNSEN, H. (1982). Electrophysiology of mammalian thalamic neurones *in vitro*. *Nature* **297**, 406–408.

LLINÁS, R. & MÜHLETHALER, M. (1988). An electrophysiological study of the *in vitro*, perfused brain stem–cerebellum of adult guinea-pig. *Journal of Physiology* **404**, 215–240.

LLINÁS, R. & SUGIMORI, M. (1980*a*). Electrophysiological properties of *in vitro* Purkinje cell somata in mammalian cerebellar slices. *Journal of Physiology* **305**, 171–195.

LLINÁS, R. & SUGIMORI, M. (1980*b*). Electrophysiological properties of *in vitro* Purkinje cell dendrites in mammalian cerebellar slices. *Journal of Physiology* **305**, 197–213.

LLINÁS, R. & VOLKIND, R. A. (1973). The olivo-cerebellar system: functional properties as revealed by harmaline-induced tremor. *Experimental Brain Research* **18**, 69–87.

LLINÁS, R. & YAROM, Y. (1981*a*). Electrophysiology of mammalian inferior olivary neurones *in vitro*. Different types of voltage-dependent ionic conductances.. *Journal of Physiology* **315**, 549–567.

LLINÁS, R. & YAROM, Y. (1981*b*). Properties and distribution of ionic conductances generating electroresponsiveness of mammalian inferior olivary neurones *in vitro*. *Journal of Physiology* **315**, 569–584.

LLINÁS, R. & YAROM, Y. (1986). Oscillatory properties of guinea-pig olivary neurones and their pharmacological modulation: an *in vitro* study. *Journal of Physiology* **376**, 163–182.

MARR, D. (1969). A theory of cerebellar cortex. *Journal of Physiology* **202**, 437–470.

MUGNAINI, E. & OERTEL, W. H. (1981). Distribution of glutamate decarboxylase positive neurons in the rat cerebellar nuclei. *Society for Neuroscience Abstracts* **7**, 122.

MURASE, K. & RANDIC, M. (1983). Electrophysiological properties of rat spinal dorsal horn neurones *in vitro*: calcium-dependent action potentials. *Journal of Physiology* **334**, 141–153.

NOWYCKY, M. C., FOX, A. P. & TSIEN, R. W. (1985). Three types of neuronal calcium channel with different calcium agonist sensitivity. *Nature* **316**, 440–443.

SASAKI, K. & LLINÁS, R. (1985). Evidence for dynamic electrotonic coupling in mammalian inferior olive *in vivo*. *Society for Neuroscience Abstracts* **11**, 181.

SOTELO, C., GOTOW, T. & WASSEF, M. (1986). Localization of glutamic acid-decarboxylase-immunoreactive axon terminals in the inferior olive of the rat, with special emphasis on anatomical relations between GABAergic synapses and dendrodendritic gap junctions. *Journal of Comparative Neurology* **252**, 32–50.

TAKEUCHI, Y., KIMURA, H. & SANO, Y. (1982). Immunohistochemical demonstration of serotonin-containing nerve fibers in the cerebellum. *Cell Tissue Research* **226**, 1–12.